

Comparison between methods for simulations

We need to contrast our results with other methods and demonstrate that we are learning something useful. Since there are no direct comparison methods, we can't do the usual comparison of TPR and FPR. We need to be creative.

I like the results as you summarize them in Fig 4, but I think we also need a way to compare our finding with those of the other competing methods.

The challenge with summarizing our results is that we have a **3D realization**: all pairs of perturbations, and which programs. We could illustrate this as a tensor. Imagining myself as a scientist, I might see that A & B converge (pvalue small) and then look to see which program(s) they converge on. The display in Fig 4 captures much, but not all of this information, which is ok. But if we want to compare with other methods, we could numerically summarize our success by counting in the tensor how many times we capture the true signal and also how many times we capture false signals. We have a choice here: we could chose to count nothing, if the overall p-value is not significant; or we could do something like you are doing for the bottom portion of figure 4.

For a competing method, they have a 2D representation (perturbations by programs), but no direct comparison of perturbations. One could imagine computing a tally of success by extrapolating this representation to a tensor by doing the following. Pick a program. Consider all pairwise perturbations. We say they converge on a program if both have significant p-values. Iterate over all programs. Ultimately we can count the rate of success, as we did for our method. I expect they will have a lot of **false positives**.

I think two methods are the possible comparisons for us b/c they do a significance test (schnitzler2024 and yang2025). It really is very hard to do comparisons because these methods have different ways of computing programs, making it impossible to compare. I think we have to simply choose a method for programs and use it for every method.

What to display results for the Yao analysis using other methods?

A potential advantage of our method is that it accounts for multiple comparisons internally. Hopefully we get more interpretable results. Let us compare methods on Yao data. These data seem to show a lot of convergence with our method. Maybe we can show the other methods find so much it is not interpretable.

Consider the following summary to compare with our Fig 5a.

- 1) Start with the perturbation by program display of p-values or fold changes like Fig 4a using whatever method they recommend for p-values.
- 2) Create a perturbation by perturbation upper triangular by providing a gold dot every time the 2 perturbations are significant for at least one program. (Hopefully this will be dense and uninterpretable).
- 3) In the lower triangular of this matrix, tally how many programs had common signals.

Now some ideas for how to capitalize on the results using our method.

The idea now is that we need to provide something interesting for scientist from the Yao results.

For the Suppl generate all the (convergence x program) results with at least 4 convergent elements and make igraphs.

Identify every convergent pair with at least 4 convergent programs. Do the set of programs have similar GO terms, or do the union of elements have interesting GO terms? It would be nice to say something about these highly convergent pairs.